

Oxehealth heartbeat monitor

Oxehealth is a start up that plans to use cameras to monitor your heartbeat from across the room. <http://dgit.in/Oxealth>



T-Mobile reveals "Digits"

This lets customers use one common phone number connected across all their other devices. <http://dgit.in/TmobileDgts>

residents- 600 at maximum capacity. Also, the entire unit can recharge completely with only seven hours of straight sunlight exposure.

Since Ta'u is pretty remote, the citizens of the island were very much used to frequent power rationing, outages and also a high dependency on diesel generators. The core services of the island include schools, the local hospital and police station. Thanks to the installation of the micro-grid, they don't have to worry about power rationing or outages anymore.

Aside from bringing energy to the islanders, the project also enables the island to save enormously on energy costs and also offset the use of over 109,500 gallons of diesel a year.

Also, to the credit of the two companies involved, Tesla and SolarCity's Ta'u microgrid project was put together from start to finish within just a year. And with the project, almost 100 percent of the island's energy requirements are met by renewable sources.

How valid is the project as a proof of concept?

There exists no doubt that the Ta'u project is a landmark project in the journey of using the sun as a source of energy. But how valid is it as a proof of concept?

To find the viability of a solar energy project, five aspects are to be looked into before start:

1. The cost of setting up the plant
2. What it means to be economically viable
3. The solar power that's expected to be generated
4. The required solar power for the citizenry
5. The environmental impact

1. The cost of setting up the plant

One of the reasons why solar power plants are being erected in more number than ever before is the fall in price of silicon over the past half a decade or so. As per a conservative estimate, putting up a 100 kw plant would cost just over a crore in Indian rupees.

The Solar City project has a solar generation capacity of 1.4 megawatts

(the 6 megawatts of battery storage capacity is in addition to this).

This means that the cost for the project of this size would fall in the range of ₹14 to 15 Cr.

2. Economic viability

Due to the generally higher rate of inflation in India as opposed to developed countries like the USA, 15% is considered to be an acceptable rate of return for the country (for a developed nation, that could be 8%).

However, in addition to the return, one should also take into account the Operation & Maintenance cost – which usually falls in the range of 1% of the project cost.

Therefore, the return (expected) + O&M cost = 16% (In India) and 9%(in developed countries).

As for the Ta'u project the factors such as the exact return expected are not yet available for public consumption. However, confidence-inducing statements have come from Tesla to its stakeholders. Also, the benefit of the project is already being enjoyed by the islanders. This success would moreover be viewed as a way to attract more potential clients for Tesla. In that sense, the return expected is seriously long-term and there's every reason to assume that they are justified in it.

3. Expected solar power generation

The amount of solar energy generated is dependent on a few factors- like the ambient temperature, radiation and the technology that is used. Out of these, temperature and radiation are in turn location-dependent. It's said that a 95% accurate generation estimate is now possible thanks to sophisticated meteorological databases that are available.

For instance, a 100 kw solar power plant in the USA would give be able to generate 1,76,804 units while in India, the same plant would create 1,78, 321 units every year.

The Ta'u island is located in a part of the world where the sunlight is ample. This means it's safe enough to estimate

that the number of units produced are comparatively higher than in most of the North American regions.

4. The required solar power of the citizenry

The solar project is already fulfilling almost the complete power requirements of the island. This Tesla and Solar City has already proven. This means that the energy created by the plant is sufficient for the citizenry.

5. The environmental impact

Apparently, a "huge" amount of forest land had to be cleared to set up the plant (The exact area mowed down is still not easily available with varied information coming from different sources). However, thanks to the project, more than 4 lakh liters of diesel would be offset every year– the island was previously dependent on diesel to meet their power requirement.

While factors such a rising fuel cost and the cost of transporting such huge quantities to the island are taken in to consideration, the long term financial gains are significant.

Other concerns

One of the major peripheral concerns would be the power generation capabilities in adverse weather conditions. The storage capacity for the project comes around 6 MW–which by even the most extreme estimates would suffice for the island in a situation when they wouldn't have ample sunlight even for prolonged days.

Final analysis

The project has the advantage of a location that gets ample sunlight and which has been reliant on fossil fuel, the offsetting of which in itself makes the project viable.

The environmental cost could be justified in this instance due to the aforementioned reason. And that is one point in which the critics may find the project less than adaptable universally – the justification for clearing huge swathes of land may not exist in all kinds of places.